

Project	AGAGO 132/33kV Substation, the 132kV Extension Bays at Gulu Substation and Agago HPP 132kV Switchyard, and the Associated Works	Designed	RK
System	132 kV System	Checked	MM
Title	Rigid Conductor Ampacity Calculation	Approved	DM

1 PURPOSE

The purpose of the document is to select optimum sizing of Tubular Conductor

2 REFERENCE STANDARDS

IEC - 61597 (1995) Overhead electrical conductors – Calculation methods for stranded bare conductors

IEC - 60865 (1993) Short Circuit Current calculation of Effects

IEEE 605:2010 IEEE Guide for Bus Design in Air Insulated Substations

3 SYMBOLS & UNITS

Symbol	Description	Unit Representations
α	Temperature coefficient of electrical resistance at temperature T_a	$^{\circ}\text{C}^{-1}$
A_s	Cross sectional area of conductor	Sq.mm
a_s	Sub Conductor spacing	m
a	Phase to phase spacing of conductor	m
C	Specific heat capacity	$\text{Jkg}^{-1}\text{^{\circ}C}^{-1}$
d_o	Outer diameter of the tube	m
d_i	Inner diameter of the tube	m
K_e	Emissivity constant with respect to black body	
f	Frequency	Hz
F	Factor for Skin effect Calculation	
h	Height of conductor above ground	m
k_{20}	Specific conductivity	$\Omega^{-1}\text{m}^{-1}$
P_{conv}	Convection heat Loss	Wm^{-1}
P_{rad}	Heat loss by radiation of the conductor	Wm^{-1}
P_{sol}	Solar heat gain by the conductor surface	Wm^{-1}
S_i	Intensity of Solar radiation	Wm^{-2}
ρ	Density	kgm^{-3}
T_1	Ambient Temperature of air	$^{\circ}\text{C}$
T_{af}	Conductor temperature after fault	$^{\circ}\text{C}$
T_{bf}	Conductor temperature before fault	$^{\circ}\text{C}$
T_{fd}	Duration of fault current	s
V	System Voltage	kV



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v	Wind Speed at Still air condition	ms^{-1}
I_{sc}	Fault level	kA
I_L	Load current	A
A	Altitude (Substation Elevation) above mean sea level	m
m	Conductor irregularity factor	
T_0	Temperature used by IEEE to develop empirical constant	$^{\circ}\text{C}$
D_a	Relative air density factor (is a function of altitude and temperature)	
r_c	Conductor outside radius	cm
E_c	Calculated corona onset gradient	kVcm^{-1}
T_2	Final equilibrium temperature	$^{\circ}\text{C}$
r_{dc}	DC Resistance at 20°C	ohm Km^{-1}
t	Thickness of Conductor	m
E_{air}	Disruptive gradient of air	kVm^{-1}
γ	Solar absorption coefficient	
S	Stefan's Constant	$\text{Wm}^{-2}\text{K}^{-4}$
λ	Thermal conductivity of the air film in contact with conductor	$\text{Wm}^{-1}\text{K}^{-1}$
P_j	Heat Generated by Joule effect	Wm^{-1}
Nu	Nusselt number	
Re	Reynolds number	
I_{max}	Continuous Current carrying capacity	A
K_{sc}	Factor for calculating Short time withstand current	
S_{thr}	The Rated Short Time Withstand Current Density	Am^{-2}
n	No of Runs	
I_{scc}	Short-time withstand Current	kA
V_1	Line to ground voltage	kV
g	Factor for subconductors	
C"	Empirical constant	cm^{-1}
P / P_0	Ratio for calculating relative air density factor	
r_e	Equivalent single conductor radius of bundle subconductors	cm
h_e	Equivalent distance from center of Conductor to ground plane for three phase	cm
E_a	Average voltage gradient at the surface of the conductor	kVcm^{-1}
E_m	Maximum voltage gradient at the surface of the conductor	kVcm^{-1}
r_t	AC resistance at given temperature	ohm m^{-1}



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4 SYSTEM PARAMETERS

4.1	System Voltage	V	=	132	kV	As per specification
4.2	Frequency	f	=	50	Hz	As per specification
4.3	Fault level	I_{sc}	=	31.5	kA	As per specification
4.4	Duration of fault current	T_{fd}	=	1	sec	As per specification

5 ENVIRONMENTAL DETAILS

5.1	Wind Speed at still Air Condition	v	=	2.5722	ms^{-1}	Volume 2/2: Technical Specifications , 2.1 SITE AND SERVICE CONDITIONS
5.2	Ambient Temperature of air	T_1	=	37.2	°C	Volume 2/2: Technical Specifications , 2.1 SITE AND SERVICE CONDITIONS
5.3	Intensity of Solar radiation	S_i	=	437	Wm^{-2}	Considered 437 W/m2 as per Annexure-2

6 LAYOUT DETAILS

6.1	Height of conductor above ground	h	=	4.9	m	As per Layout
6.2	Phase to Phase spacing of conductor	a	=	3	m	As per Layout

7 CONDUCTOR DATA

7.1	Material		=	Aluminium		
7.2	Category		=	TUBE		
7.3	Sub category		=	IPS		
7.4	Code name of Conductor		=	2-1/2'		
7.5	DC Resistance @ 20° C	r_{dc}	=	3.20000E-05	ohm Km^{-1}	Annexure 1:conductor data
7.6	Cross sectional area of conductor	A_s	=	0.001018	sq.mm	Annexure 1:conductor data
7.7	Outer diameter	d_o	=	0.073	m	Annexure 1:conductor data
7.8	Inner diameter	d_i	=	0.0635	m	Annexure 1:conductor data



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7.9	Conductor temperature after fault	T_{af}	=	200	°C	IEC 60865-1 (table 6)
7.10	Conductor temperature before fault	T_{bf}	=	67.2	°C	As per Specification
7.11	Thickness of Conductor	t	=	0.00475	m	
7.12	Final equilibrium temperature	T_2	=	67.2	°C	As per Specification

8 OTHER DATA

8.1	Disruptive gradient of air	E_{air}	=	21.1	kvc m^{-1}	IEEE 605 : 2008 D.1.1 Corona physical process
8.2	Emissivity with respect to black body	K_e	=	0.6		IEC 61597(clause 3.10)
8.3	Solar absorption coefficient	γ	=	0.5		IEC 61597(clause 3.10)
8.4	Temperature coefficient of electrical resistance at temperature T_a	α	=	0.00403	K^{-1}	IEC 61597(clause 3.10)
8.5	Specific conductivity	k_{20}	=	3.48E07	$\Omega^{-1}m^{-1}$	IEC60865-1(clause A.11)
8.6	Density	ρ	=	2700	kg m^{-3}	IEC60865-1(clause A.11)
8.7	Specific heat capacity	C	=	910	Jkg $^{-1}^{\circ}C^{-1}$	IEC60865-1(clause A.11)
8.8	Stefan's Constant	S	=	5.67E-08	W $m^{-2}K^{-4}$	IEC 61597(clause 3.6)
8.9	Thermal conductivity of the air film in contact with conductor	λ	=	0.02585	W $m^{-1}K^{-1}$	IEC 61597(clause 3.6)

9 CALCULATION - PRELIMINARY

Temperature effect on resistance

Resistance at 67.2° C

$$\begin{aligned}
 r_{dc} &= r_{dc}(1 + \alpha(T_{bf} - 20)) && \text{IEC 61597(clause. 4.2)} \\
 &= 3.20E-05 * (1 + 0.00403 * (67.2 - 20)) \\
 &= 3.8087E-05 && \text{ohm } m^{-1}
 \end{aligned}$$

Factor for calculating skin effect coefficient

$$X = \sqrt{f / r_{dc}}$$

Where Rdc in ohm per feet

$$r_{dc} = 1.161E-05 \quad \text{ohm/ft}$$

$$X = \sqrt{50 / 1.161E-05}$$

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		=	65.63	
Ratio of Thickness to Diameter	t/d_o	=	0.065068	
Skin effect coefficient for the Factor	F	=	1.00	
AC resistance	r_t	=	3.809E-05	ohm m ⁻¹

10 CONDUCTOR SIZING

10.1 HEAT BALANCE EQUATION

Under thermal Equilibrium

$$P_j + P_{sol} = P_{rad} + P_{conv} \quad \text{IEC 61597 (clause 3.2)}$$

where

P_j	=	Power losses due to Joule effect	Wm ⁻¹
P_{sol}	=	Solar heat gain	Wm ⁻¹
P_{rad}	=	Heat loss by radiation	Wm ⁻¹
P_{conv}	=	Forced convection heat loss	Wm ⁻¹

10.2 CONVECTION HEAT LOSS

$$P_{conv} = \lambda Nu (T_2 - T_1) \pi \quad \text{IEC 61597 (clause 3.7)}$$

where Nu is the Nusselt number

$$Nu = 0.65 R_e^{0.2} + 0.23 R_e^{0.61}$$

Re is the Reynolds number

$$\begin{aligned} Re &= 1.644 * 10^{09} * v * d [T_1 + 273 + 0.5(T_2 - T_1)]^{-1.78} \\ &= 1.644 * 10^{09} * 2.5722 * 0.073 * [37.2 + 273 + 0.5 * (67.2 - 37.2)]^{-1.78} \\ &= 10420.81 \\ Nu &= 69.10 \end{aligned}$$

$$P_{conv} = 168.25 \quad \text{Wm}^{-1}$$

10.3 RADIATION HEAT LOSS

$$\begin{aligned} P_{rad} &= S \pi d K_e [(T_2 + 273)^4 - (T_1 + 273)^4] \quad \text{IEC 61597 (clause 3.6)} \\ &= 5.670E-08 * \pi * 0.073 * 0.6 * [(67.2 + 273)^4 - (37.2 + 273)^4] \\ &= 32.25 \quad \text{Wm}^{-1} \end{aligned}$$

10.4 Solar Heat Gain

$$\begin{aligned} P_{sol} &= \lambda * S_i * d \quad \text{IEC 61597 (clause 3.5)} \\ &= 0.5 * 437 * 0.073 \\ &= 15.95 \quad \text{Wm}^{-1} \end{aligned}$$

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10.5 POWER LOSS DUE TO JOULE EFFECT

$$P_j = r_t I_{\max}^2 \quad \text{IEC 61597 (clause 3.4)}$$

10.6 CALCULATION OF CONTINUOUS CURRENT CAPACITY

From heat balance equation

$$I_{\max}^2 R_T + P_{\text{sol}} = P_{\text{conv}} + P_{\text{rad}} \quad \text{IEC 61597 (clause 3.8)}$$

$$I_{\max} = \sqrt{((P_{\text{rad}} + P_{\text{conv}} - P_{\text{sol}}) / R_T)}$$

$$I_{\max} = \sqrt{((32.25) + (168.25) - (15.95)) / (3.81 \times 10^{-5})}$$

$$= \boxed{2201.26} \quad \text{A}$$

10.7 SC AMPACITY

IEC 60865-1 (2011) - Short-circuit currents – Calculation of effects

$$K_{\text{sc}} = \sqrt{(k_{20} C \rho / \alpha_{20}) \ln [1 + \alpha_{20}(\theta_e - 20) / 1 + \alpha_{20}(\theta_b - 20)]}$$

$$= \sqrt{(3.480 \times 10^7 * 910 * 2700 / 0.00403) \ln [1 + 0.00403 * (200 - 20) / 1 + 0.00403 * (67.2 - 20)]}$$

$$= \boxed{88759786.37}$$

The Rated Short Time Withstand Current Density

$$S_{\text{thr}} = K_{\text{sc}} / \sqrt{T_{\text{fd}}}$$

$$= 88759786.366 / 1$$

$$= \boxed{88759786.37} \quad \text{Am}^{-2}$$

11 CALCULATIONS FOR RIGID BUS

11.1 Load current I_L $\boxed{2000}$ A

11.2 Selected No. of runs n $= \boxed{1}$

11.3 Continuous current rating of selected No. of runs $= I_{\max} * \text{Selected No of Runs}$

$$= 2201.265 * 1$$

$$= \boxed{2201.26} \quad \text{A} \quad > 2000 \text{ A}$$

11.4 Short-time withstand Current I_{scc} $= S_{\text{thr}} * A_s * n$

$$= 88759786.37 * 0.001018 * 1$$

$$= \boxed{90.36} \quad \text{kA} \quad > 31.5 \text{ kA}$$



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12 CORONA CHECK CALCULATION FOR RIGID BUS

Condition for acceptable design $E_m < E_c$

Altitude (Substation Elevation) above mean sea level

$$A = 1100 \text{ m}$$

Volume 2/2: Technical Specifications , 2.1 SITE AND SERVICE CONDITION (Gul

Conductor outside radius (d/12)

$$r_c = 3.650 \text{ cm}$$

Line to ground voltage(110% of nominal operating line to ground voltage)

$$V_1 = 83.83 \text{ kV}$$

Factor (g=1 for bundle of 1,2,3 subconductors; g=1.12 for bundle of 4 and above subconductors)

$$g = 1$$

Empirical Constant

$$C'' = 0.301 \text{ cm}^{-1}$$

IEEE 605 : 2008 D.1.1 Corona physical process

Conductor irregularity factor

$$m = 0.85$$

IEEE 605 : 2008 D.1.1 Corona physical process

Temperature used by peek to develop empirical constant

$$T_o = 25 \text{ }^{\circ}\text{C}$$

IEEE 605 : 2008 D.1.1 Corona physical process

12.1 Ratio for calculating relative air density factor

$$\begin{aligned} P/P_o &= 1 - A / 10 \\ &= 1 - (1.1 / 10) \\ &= 0.89 \end{aligned}$$

12.2 Relative air density factor (is a function of altitude and temperature)

$$\begin{aligned} D_a &= (273 + T_o)P / (273 + T_a)P_o \\ &= (273 + 25) * (0.89) / (273 + 37.2) \\ &= 0.85 \end{aligned}$$

IEEE-605 equation(D.2)

12.3 Equivalent single conductor radius of bundle subconductors

$$\begin{aligned} r_e &= r_c (g * a_s / r_c)^{(n-1)/n} \\ &= 3.650 (1 * 0 / 3.650)^{(1-1)/1} \\ &= 3.650 \text{ cm} \end{aligned}$$

12.4 Calculated Corona onset gradient

$$\begin{aligned} E_c &= m * E_o * D_a * (1 + c'' / \sqrt{D_a * r_e}) \\ &= 0.85 * 21.1 * 0.85 * (1 + 0.301 / \sqrt{0.85 * 3.650}) \\ &= 17.85 \text{ kvcm}^{-1} \end{aligned}$$

IEEE-605 equation(D.1)

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12.5 Equivalent distance from center of Conductor to ground plane for three phase

$$h_e = h * D / \sqrt{4 * h^2 + D^2} \quad \text{cm} \quad \text{IEEE-605 (equation D.8)}$$

$$= 490.0 * 300 / \sqrt{4 * 490.0^2 + 300^2}$$

$$= \boxed{143.43} \quad \text{cm}$$

12.6 Average voltage gradient at the surface of the conductor

$$E_a = V_1 / (n * r_c) * \ln(2 * h_e / r_e) \quad \text{kvc}^{-1} \quad \text{IEEE-605 (equation D.12)}$$

$$= 83.83 / (1 * 3.650) * \ln(2 * 143.43 / 3.650)$$

$$= \boxed{5.26} \quad \text{kvc}^{-1}$$

12.7 Maximum voltage gradient at the surface of the conductor

$$E_m = h_e * E_a / h_e - r_e \quad \text{IEEE-605 (equation D.13)}$$

$$= 143.43 * 5.26 / (143.43 - 3.650)$$

$$= \boxed{5.400} \quad \text{kvc}^{-1}$$

13 RESULTS

13.1	Material	=	Aluminium	
13.2	Category	=	TUBE	
13.3	Sub Category	=	IPS	
13.4	Code name of Conductor	=	2-1/2'	
13.5	Continuous Current carrying capacity	=	2201.26 A	> 2000 A
13.6	Selected No. of runs	=	1	
13.7	Short-time withstand Current	=	90.36 kA	> 31.5 kA
13.8	Duration	=	1 Sec	
13.9	Maximum surface voltage gradient	=	5.40 kvc ⁻¹	< 17.85kvc ⁻¹